When genetic time and calendar time disagree

A guided investigation of Pascall et al. (2020), BTV-8. Work through five blocks of `Pascall_student.R`. You do not need to write functions. Predict first, run the block, then explain what the output supports.

> **The question.** Do samples from two separated outbreaks follow one continuous relationship between genetic divergence and calendar time?

The paper studied the 2006–2010 and 2015–2018 outbreaks. In the supplied teaching dataset, the sampled years are **2006–2009 and 2015–2018**. Keep the historical study separate from the current epidemiological situation.

Start from observations

```
Pascall_data <- read.csv("data/metadata.csv")
Pascall_alignment <- ape::read.dna(
  "data/alignment.fasta", format="fasta")
Pascall_ML <- ape::read.tree("data/ml_F84_I_G_164.nwk")
Pascall_seqlen <- ncol(Pascall_alignment)
# Run the complete BLOCK 1 in the script.
```

Inspect sample IDs, countries and the three date columns. `Date_av` is the supplied representative date; it is not always an exact collection date. The original bounds are retained.

> **Input audit:** the teaching alignment contains nine segments (1–7, 9, 10), not all ten. One original upper date bound is invalid; the core analysis does not use that bound. Nothing is silently repaired.

Predict → run → explain

What would you expect to see during an interval with no sampled sequences? Explain why a sampling histogram is not an epidemic curve and why a missing sampled year does not establish absence of infection.

Source: Pascall et al., PLOS Biology 18(4):e3000673, pp. 1–4; supplied metadata and original practical, pp. 1–3. Read Installation_EN.pdf before class.

Orient the tree; do not date it yet

The maximum-likelihood tree is supplied so that the session can focus on interpretation. In the normal route, you calculate the root yourself. The topology and evolutionary distances are not re-estimated from sequences in this block.

```
Pascall_unrooted <- ape::unroot(Pascall_ML)
roots <- list(
  rtt = ape::rtt(Pascall_unrooted,
    tip.dates=Pascall_sampling_time[Pascall_unrooted$tip.label],
    objective="correlation", opt.tol=1e-10, ncpu=1),
  midpoint = phytools::midpoint_root(Pascall_ML)
)
rooting <- "rtt"
Pascall_tree <- ape::ladderize(roots[[rooting]])
```

Read the output

The symbols distinguish sampling periods: open circles for 2006–2009, triangles for 2015–2018. The horizontal distance is still measured in **substitutions per site**, not years. Tip labels are hidden in the overview; identifiers remain in the data objects.

> **Two root hypotheses.** The `rtt` option searches for a root giving a strong positive distance–date correlation. Midpoint rooting places the root halfway along the longest tip-to-tip path. Neither option turns the tree into a transmission tree.

Predict → run → explain

Where do the later samples fall? Are they scattered throughout the tree or grouped together? Which quantities should stay unchanged if only the root is moved?

One controlled comparison

Change `rooting` to `"midpoint"`, rerun blocks 2–3 and compare the temporal plots. Return to `"rtt"` before the main dating comparison. New output is written into the current run folder.

> The original practical used all sampling dates for rooting; this exercise preserves that choice and makes its sensitivity visible. The article used a root estimated from the first outbreak. This is an adaptation, not an exact replication.

Sources: original practical, pp. 3–5; article, pp. 11–12; ape documentation for rtt.

One line can hide two histories

Before fitting a time tree, inspect the connection between sample dates and evolutionary distances. Calculate these distances from the rooted **genetic** tree, not from an already dated tree.

```
year <- Pascall_sampling_time[Pascall_tree$tip.label]
distance <- ape::node.depth.edgelength(Pascall_tree)[
  seq_len(ape::Ntip(Pascall_tree))]
rtt_data <- data.frame(year=unname(year),
  distance=distance, outbreak=cohort)
fit_all <- lm(distance ~ year, data=rtt_data)
fit_first <- lm(distance ~ year, data=rtt_data,
  subset=outbreak=="first")
fit_second <- lm(distance ~ year, data=rtt_data,
  subset=outbreak=="second")
```

Compare two views

First inspect the pooled line; then display the two within-outbreak fits. Their slopes describe accumulation of divergence per year. Compare their vertical positions as well as their slopes.

> **Prediction:** under continuous evolution at a similar rate, where should the second outbreak lie relative to the extrapolated first-outbreak trend?

Questions to discuss

Does a good pooled R^2 guarantee that a single clock describes the entire history? Can divergence increase within both outbreaks while the interval between them is inconsistent with the same continuous process?

> **Descriptive, not independent tests.** Tips share ancestry. Do not interpret the ordinary OLS p-value or confidence interval as independent phylogenetic evidence. These OLS lines are not the gamma regression and Bayesian analysis used in the article.

Repeat this step under the alternative root. Which aspect of the pattern is robust and which is sensitive? An extrapolated intercept is not a directly observed date, and not automatically the date of an ancestor.

Sources: article, Fig. 4B and caption (p. 6), downstream statistical analysis (p. 12); explicit descriptive extension to the original practical.

Fit two clock models explicitly

Now estimate node times under two assumptions. Both fits receive the same topology, root, sampling dates and alignment length. Keep warnings visible: successful computation does not imply that the model describes the data well.

```
Pascall_dater_strict <- treedater::dater(
  tre=Pascall_tree,
  sts=Pascall_sampling_time[Pascall_tree$tip.label],
  s=Pascall_seqlen, clock="strict",
  minblen=1/365.25, temporalConstraints=TRUE,
  numStartConditions=3, ncpu=1, quiet=FALSE)
Pascall_dater_relaxed <- treedater::dater(
  tre=Pascall_tree,
  sts=Pascall_sampling_time[Pascall_tree$tip.label],
  s=Pascall_seqlen, clock="uncorrelated",
  minblen=1/365.25, temporalConstraints=TRUE,
  numStartConditions=3, ncpu=1, quiet=FALSE)
```

Read the estimated quantities

Dates are decimal years, and rates are substitutions per site per year. The strict clock shares one rate across branches; the relaxed model permits rate variation. The one-day minimum branch duration is a numerical setting, **not** a claim about sampling precision or infection timing.

Predict → run → explain

Which parts of the dated tree change when rate variation is allowed? How do the estimated root date and mean rate compare? Why is extra flexibility not itself proof of a biological explanation?

> **No automatic rescue.** If R fails, retain the error and use the explicit rescue route. The browser shows the published BEAST figure as a reference; it does not run treedater. The two methods must not be conflated.

Sources: original practical, pp. 6–8; treedater documentation. The article used a partitioned BEAST analysis with a lognormal relaxed clock, not these treedater point estimates.

Find the branch, then qualify the claim

Find the common ancestor of the later samples by their IDs, not by a hard-coded node number. Identify the branch entering that clade, then examine its estimated rate.

```
td <- Pascall_dater_relaxed
late_ids <- Pascall_data$ID[Pascall_data$outbreak=="second"]
late_node <- ape::getMRCA(td, late_ids)
stem_edge <- which(td$edge[,2] == late_node)
branch_rates <- td$omegas
phytools::plotBranchbyTrait(plain_phylo(td),
  log10(branch_rates), mode="edges",
  show.tip.label=FALSE, legend=TRUE, prompt=FALSE,
  title="log10 rate: substitutions/site/year")
```

> `omegas` contains fitted branch rates; `edge.p` contains diagnostic cumulative probabilities. They are not interchangeable. Never reorder the tree edges without reordering their associated values.

Three levels of conclusion

Level	Your statement
Observation	Describe the pattern in the sequence-derived distances and sampling dates.
Inference	State what a specified clock model implies, conditional on the data, root and assumptions.
Hypothesis	Suggest an explanation without claiming that these data identify its exact source.

Compare your output with the paper's Fig. 3. A mismatch deserves investigation; do not change settings merely to force agreement. The published argument concerns unusually little evolutionary change between outbreaks. The authors explicitly state that the actual source of reintroduction was not identified.

Final response

Write one supported finding, one important limitation and one additional type of evidence needed to distinguish explanations.

Optional, instructor-led: fixed-root date sensitivity; sampled lineages through time; a parametric bootstrap. A lineage count is not incidence, prevalence or effective population size.

Sources: article, pp. 5–9; original practical, pp. 9–12; treedater source and phytools documentation. Complete citations: SOURCES.md.